



Digital Twin And Its Applications In The Internet World

Digital twin is a cloud-based virtual model of a process, product or service. Billions of things will be represented by digital twins in the next five years, showing the tremendous potential of this technology



Vinayak Ramachandra Adkoli is BE in industrial production. He has been a lecturer in mechanical department for ten years, in three different polytechnics. He is also a freelance writer and cartoonist

A digital twin is a digital replica of a living or non-living physical entity. Digital twin refers to a digital replica of physical assets (physical twin), processes, people, places, systems and devices that can be used for various purposes. Digital twin is a real mapping of all components in a product life cycle using physical data, virtual data and interaction between these data.

Digital twins integrate the Internet of Things (IoT), artificial intelligence (AI), machine learning (ML) and software analytics with special network graphs to create living digital simulation models. These

models update and change as their physical counterparts change. A digital twin also integrates historical data from past machine usage into its digital model.

In various industrial sectors, twins are being used to optimise operations and for maintenance of physical assets, systems and manufacturing processes. Digital twins are also called Cyber Objects or Digital Avatars.

Digital twin to improve several sectors

Digital twin technology finds applications in several sectors such as energy and utilities, aerospace and defence, machines manu-